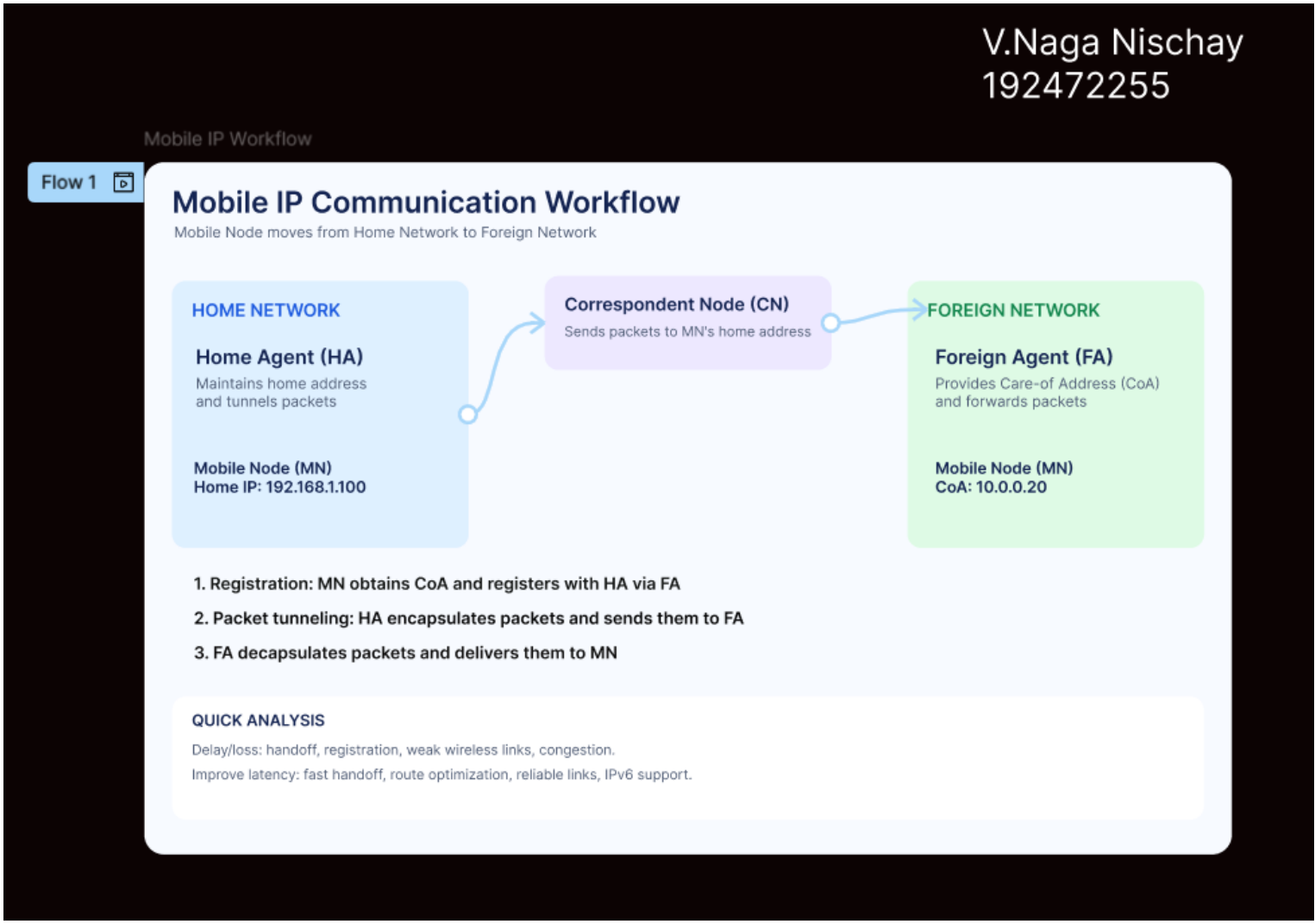
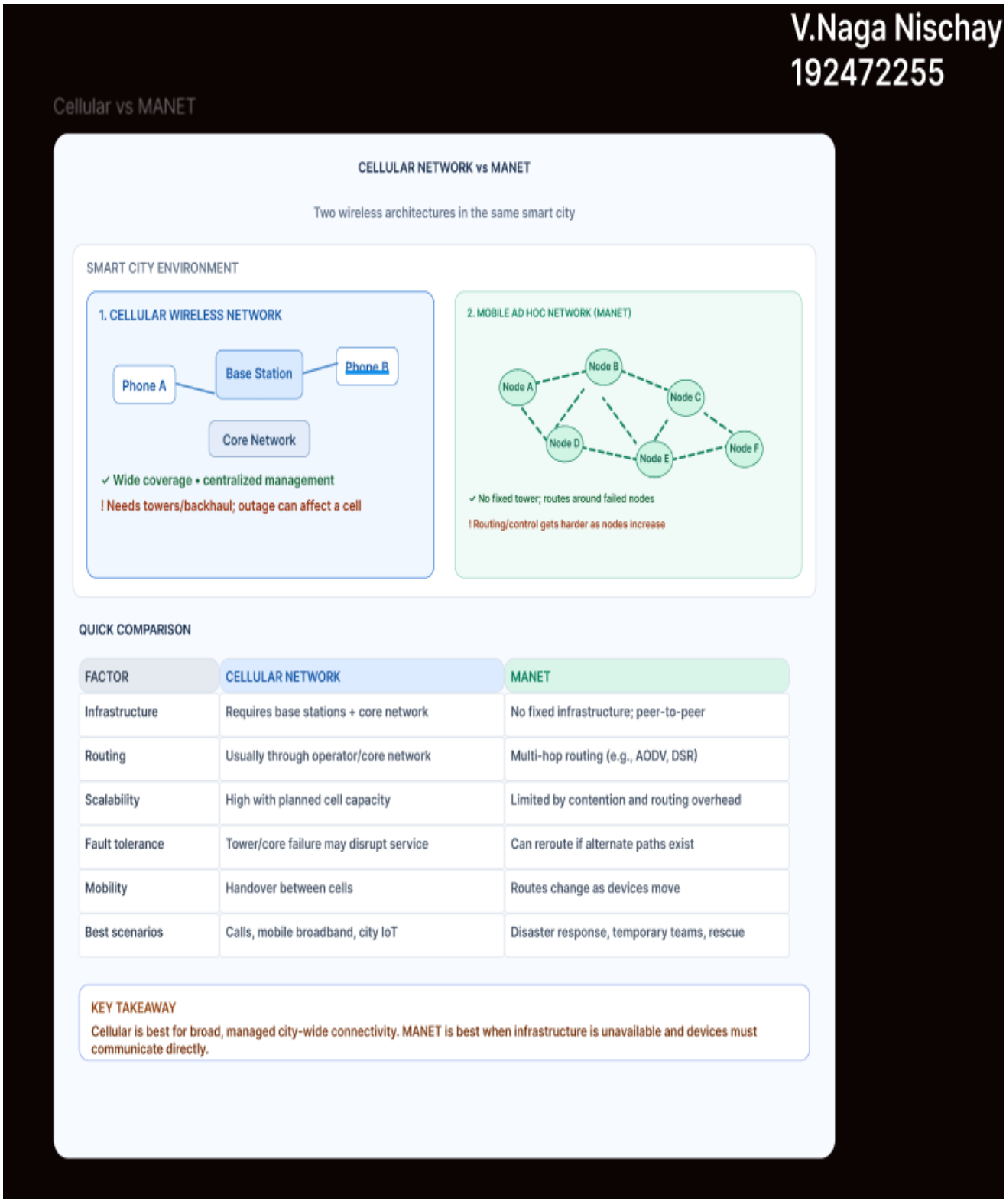
1)


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Mobile IP Workflow
Flow 1
Mobile IP Communication Workflow
Mobile Node moves from Home Network to Foreign Network
HOME NETWORK
Home Agent (HA)
Maintains home address and tunnels packets
Mobile Node (MN)
Home IP: 192.168.1.100
Correspondent Node (CN)
Sends packets to MN's home address
FOREIGN NETWORK
Foreign Agent (FA)
Provides Care-of Address (CoA) and forwards packets
Mobile Node (MN)
CoA: 10.0.0.20
1. Registration: MN obtains CoA and registers with HA via FA
2. Packet tunneling: HA encapsulates packets and sends them to FA
3. FA decapsulates packets and delivers them to MN
QUICK ANALYSIS
Delay/loss: handoff, registration, weak wireless links, congestion.
Improve latency: fast handoff, route optimization, reliable links, IPv6 support.

2)


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Cellular vs MANET
CELLULAR NETWORK vs MANET
Two wireless architectures in the same smart city
SMART CITY ENVIRONMENT
1. CELLULAR WIRELESS NETWORK
Phone A
Base Station
Phone B
Core Network
✓ Wide coverage • centralized management
! Needs towers/backhaul; outage can affect a cell
2. MOBILE AD HOC NETWORK (MANET)
Node A
Node B
Node C
Node D
Node E
Node F
✓ No fixed tower; routes around failed nodes
! Routing/control gets harder as nodes increase
QUICK COMPARISON
FACTOR
CELLULAR NETWORK
MANET
Infrastructure
Requires base stations + core network
No fixed infrastructure; peer-to-peer
Routing
Usually through operator/core network
Multi-hop routing (e.g., AODV, DSR)
Scalability
High with planned cell capacity
Limited by contention and routing overhead
Fault tolerance
Tower/core failure may disrupt service
Can reroute if alternate paths exist
Mobility
Handover between cells
Routes change as devices move
Best scenarios
Calls, mobile broadband, city IoT
Disaster response, temporary teams, rescue
KEY TAKEAWAY
Cellular is best for broad, managed city-wide connectivity. MANET is best when infrastructure is unavailable and devices must communicate directly.